STUDY PROJECT PHASE-1

BCSZC241T

Explainable Machine Learning Framework for Battery Health Assessment and Dynamic Backup-Time Prediction

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1. Project Idea Summary

1.1 Abstract

Battery-powered systems such as uninterruptible power supplies, inverters, electric vehicles, communication equipment, and stationary energy storage depend on accurate, up-to-date information about battery condition and the remaining time a battery can support a connected load. In practice, backup-time estimates are usually based on fixed formulas or manufacturer ratings that do not adequately capture the effects of aging, temperature, discharge rate, and changing operating conditions. Battery State of Health (SOH) is difficult to estimate directly during normal operation, because usable capacity changes gradually over repeated charge-discharge cycles.

Research on SOH and capacity estimation, on remaining-runtime or remaining-energy prediction, and on explainable AI is individually well established, but the literature and existing tools generally treat these as separate problems rather than as parts of one integrated, user-facing workflow (Berecibar et al., 2016; Roman et al., 2021; Wang & Chen, 2020; Tu et al., 2024). This project proposes a modular software framework built around that gap. Battery measurements are imported, validated, and cleaned; discharge-cycle and load-related features are engineered from the cleaned data; and machine-learning models are trained to estimate remaining capacity, State of Health, and dynamic backup time. A transparent, configurable formula-based calculation is kept alongside the machine-learning models throughout, as a benchmark rather than as a fallback. Random Forest regression is used as the initial prediction method and is compared, under the same evaluation protocol, against a simple Linear Regression comparator so that any gain from nonlinear modelling is demonstrated rather than assumed. Feature importance, permutation importance, and SHAP explanations show why a given prediction was produced, and point predictions of capacity and backup time are accompanied by an uncertainty interval rather than a single number.

The framework is developed and evaluated using a combination of project-generated synthetic data (for pipeline and schema testing only) and independent public lithium-ion laboratory datasets, principally a second-life battery dataset, NASA ageing datasets, an Oxford battery-degradation dataset, and the CALCE battery dataset, so that performance claims rest on real experimental data rather than a single source. In line with the wider literature, the project's contribution is best described as integration rather than invention: reliable individual methods already exist for SOH estimation, runtime/energy prediction, and model explanation (Khaleghi et al., 2024; Lundberg & Lee, 2017; Tu et al., 2024), but no open, reproducible tool combines all of them, together with a transparent formula baseline and independent validation, into a single decision-support system for stationary inverter and backup-power applications. The current stage of the project (Phase 1) is limited to problem identification, dataset commitment, and planning; implementation begins in Phase 2-3.

Keywords: State of Health (SOH), Dynamic Backup-Time Prediction, Explainable Machine Learning, Random Forest Regression, SHAP(SHapley Additive exPlanations), Leakage-Safe Feature Engineering, Lithium-Ion Battery Degradation, LOBO(Leave-One-Battery-Out) Cross-Validation, Predictive Maintenance.

2. Project Background, Problem Statement, and Motivation

2.1 Background

Rechargeable batteries experience gradual degradation during use. Their available capacity, internal resistance, voltage response, thermal behaviour, and ability to support high loads change with age, temperature, discharge rate, operating history, and chemistry, so two batteries with the same nominal rating can provide different backup times under the same load. Public research datasets show that this variability is commonly studied through repeated charge, discharge, and impedance-measurement cycles at different temperatures and load levels; the NASA Ames Prognostics Center of Excellence (PCoE) datasets, for example, contain measurements from cells subjected to different charge-discharge profiles, temperatures, and current loads (Saha & Goebel, 2007), while the Oxford battery-degradation work applies a dynamic urban discharge profile to lithium-ion pouch cells to study degradation under realistic, time-varying current (Birkl, 2017).

State of Health, normally expressed as the ratio between present usable capacity and rated capacity, is a key indicator for maintenance and replacement decisions, but it cannot always be measured directly during normal operation. Likewise, actual backup time is only known with certainty after a complete discharge test, which is often impractical. Because separate strands of research have developed largely independent methods for SOH/capacity estimation and for remaining-runtime prediction (Roman et al., 2021; Wang & Chen, 2020; Tu et al., 2024), this project treats the two as interconnected decision-support needs rather than as isolated outputs, and adds a transparent baseline and explanations so that predictions can be checked and understood rather than taken on faith.

2.2 Problem Statement

The core problem is the absence of a single, reproducible software workflow that combines battery-health assessment, load-aware backup-time prediction, a transparent benchmarking formula, model evaluation, explainability, and user-oriented reporting. Existing research offers robust methods for individual components, but these are typically studied in isolation or within narrow battery-management applications (Khaleghi et al., 2024; Roman et al., 2021; Tu et al., 2024). As a result, a user may see only a battery's nominal rating, or an isolated SOH/SOC estimate, without an explainable prediction of how long the battery can support a particular load; conversely, a runtime estimate may not account for long-term degradation at all.

Users and organisations affected by this gap include:

- Households and small businesses operating inverter or UPS systems
- Data centres, communication facilities, and industrial sites that depend on backup power
- Electric-vehicle operators and maintenance teams
- Solar-energy storage and stationary energy-storage operators
- Battery service centres and maintenance engineers
- Manufacturers and researchers developing battery-management systems, and students analysing degradation data

Conventional backup-time formulas (voltage, nominal capacity, load, and a fixed efficiency value) are useful as rough baselines but do not adequately represent temperature effects, nonlinear discharge behaviour, cycle ageing, voltage sag, or battery-specific degradation. Existing battery-management solutions can also fall short because they may depend on proprietary hardware or manufacturer-specific data, provide only State of Charge rather than

long-term health, use fixed rules that do not adapt to observed degradation, produce estimates without explaining the factors behind them, address either health estimation or runtime prediction but not both, require laboratory measurements unavailable in ordinary applications, or be difficult to reproduce, extend, or demonstrate as a complete system. This project addresses that integration gap through software and machine learning; it is intended as a decision-support prototype, not a certified battery-management or safety system.

2.3 Motivation

Academic relevance. The project combines software engineering, data science, regression modelling, model evaluation, explainable AI, and human-computer interaction into one integrated workflow, going beyond a single notebook or an isolated model.

Technical relevance. Battery behaviour is nonlinear and condition dependent, requiring careful data validation, feature engineering, leakage-safe modelling, comparison against a physics-inspired baseline, reproducible evaluation, and interpretation of results.

Societal and industrial relevance. Improved battery monitoring can support reliable backup power, more efficient battery use, fewer unnecessary replacements, safer maintenance decisions, and better planning for second-life or degraded batteries. The prototype is positioned as a decision-support tool, not a safety certified controller.

Course-driven motivation. The project was selected to demonstrate that a Study Project can go beyond a report or an isolated notebook, producing a working, modular, and well-documented software prototype that explicitly states its research basis, assumptions, data sources, and limitations.

3. Educational Value and Course Alignment

3.1 Relevance to Course Objectives

Area	Project Alignment
Software engineering	Requirements analysis; modular system design; separation of data, model, evaluation, and presentation layers; error handling and input validation; reusable pipelines; version control; testing and reproducibility.
Data science and machine learning	Dataset assessment; missing-value handling; duplicate/outlier detection; feature engineering; train-test separation; regression modelling; performance evaluation and model comparison.
Artificial intelligence	Nonlinear prediction using Random Forest; model interpretation through feature importance, permutation importance, and SHAP-based global/local explanations.
Data and information management	Common schema for battery observations; validation of required fields, types, and units; structured export of processed data, model artefacts, and evaluation results.
Human-computer interaction	Interactive dashboard; dataset upload and validation feedback; user-controlled operating-condition inputs; visual comparison of model outputs; downloadable results.
Applied mathematics and statistics	Energy and backup-time equations; State-of-Health calculation; error measures (MAE, RMSE, MAPE, bias, R^2); comparison of formula-based and data-driven methods.

3.2 Expected Learning Outcomes

Technical skills: build a complete Python-based machine-learning application; use pandas and NumPy for data processing; use scikit-learn for regression modelling and evaluation; use Streamlit for the interactive dashboard; use Matplotlib and SHAP for visualisation and explainability; save and reload trained models with Joblib; work with CSV, Excel, JSON, image, and serialized model files.

Analytical skills: translate an operational problem into measurable prediction targets and input features; distinguish features from targets to avoid leakage; benchmark a formula baseline against machine-learning predictions on consistent test data; interpret prediction errors, uncertainty, and cross-dataset generalisation limits.

Design and problem-solving skills: design a modular processing pipeline and a common schema for diverse public datasets; separate health-assessment and backup-time workflows behind shared preprocessing; handle missing dependencies, changing environments, and messy or invalid data; balance technical completeness against the project's time constraints.

Documentation and communication skills: prepare proposal and design documentation; record assumptions, formulas, feature definitions, and references; maintain a structured source-code repository; present results reproducibly, clearly separating demonstration from validation.

4. Objectives

4.1 Primary Objectives

- Develop a modular software pipeline that imports, validates, cleans, and transforms battery cycling and operational data from multiple public datasets into a common schema.
- Develop a leakage-safe machine-learning workflow that estimates remaining battery capacity and calculates State of Health (SOH) relative to rated/reference capacity, using real battery datasets.
- Build a load-aware machine-learning model (Random Forest) that predicts dynamic backup time under varying load, temperature, and degradation conditions.
- Implement a transparent, configurable formula-based backup-time baseline and compare it with machine-learning predictions on the same evaluation records.
- Evaluate model performance using appropriate regression metrics (MAE, RMSE, MAPE, bias, R^2) and battery-level cross-validation as defined in Section 5.7.
- Provide interpretability through feature importance, permutation importance, and SHAP, and deliver an interactive dashboard presenting battery health, predicted backup time, comparisons, and the reasoning behind each prediction.

4.2 Secondary Objectives

- Integrate multiple independent public datasets (second-life, NASA PCoE, Oxford, CALCE) to improve robustness, generalisation, and cross-dataset validation.
- Test model generalisation across different batteries, operating conditions, temperatures, and load profiles, and record chemistry/protocol metadata for each dataset.

- Provide a rule-based, non-safety-critical recommendation layer that maps SOH, backup time, load, and data-quality indicators to maintenance guidance (inspect, reduce load, plan replacement, no action), shown alongside its supporting explanation.
- Support what-if analysis so users can vary load, temperature, or usage scenario and observe the effect on predicted SOH and backup time.
- Support CSV/Excel import and export of cleaned data, predictions, metrics, and explanation outputs; persist trained models and configuration artefacts for reuse.
- Architect the system so later phases can add algorithms, datasets, chemistries, a persistent database, or new deployment environments without a redesign, and document limitations and future extensions (including RUL prediction and field-data validation) beyond the current core scope.

5. Research and Analysis

5.1 Literature Review and Existing Approaches

5.1.1 Formula-based backup-time estimation

A transparent runtime baseline can be built from the battery's available energy and the load's power demand: Available Energy (Wh) = Average Voltage (V) × Available Capacity (Ah); Baseline Backup Time = Available Energy × Efficiency / Load Power. Equivalently, in minutes, Backup Time = (Voltage × Capacity × Efficiency / Load) × 60. Where load power is not directly available, it can be estimated from measured voltage and current. This approach is transparent and inexpensive but assumes a relatively stable load and does not directly model nonlinear degradation, temperature effects, voltage sag, cycle history, or chemistry-specific discharge behaviour (Kurucan et al., 2024). It is retained in this project as a benchmark, not as the final estimator.

5.1.2 Coulomb-counting, filter-based, and equivalent-circuit approaches

Battery-management systems frequently estimate State of Charge by integrating current over time; such methods are sensitive to current-sensor error, initial-state error, and accumulated drift, and describe present charge level rather than long-term health (Kurucan et al., 2024). More advanced state-estimation methods use observers, filters, or equivalent-circuit models: Wang and Chen (2020), for example, developed a joint framework for State of Charge and remaining discharge-time prediction using an unscented particle filter. These approaches can explicitly represent temporal battery state, but their deployment depends on model structure, parameter identification, and calibration to the specific operating conditions used, which limits reuse across chemistries and applications (Khaleghi et al., 2024).

5.1.3 Data-driven battery health estimation

Machine learning is widely used to estimate battery capacity, State of Health, and life-related indicators from measured voltage, current, temperature, and engineered features. Roman et al. (2021) presented a machine-learning pipeline for SOH/capacity-fade estimation across 179 cells, emphasising feature engineering, model calibration, and confidence bounds. Reviews by Ren and Du (2023) and Khaleghi et al. (2024) confirm that data-driven battery-state estimation is an established area, while identifying continuing challenges around data quality, operating-condition variation, and generalisation. Berecibar et al. (2016) similarly note that reliable SOH estimation remains a persistent challenge in battery management despite the range of available methods.

5.1.4 Dynamic remaining energy and backup-time prediction

Runtime is related to, but not identical with, long-term health. Tu et al. (2024) showed that remaining discharge energy depends on current rate and electro-thermal dynamics, and proposed a machine-learning approach for prediction across broad current ranges. This supports the project's decision to include operating load/current and temperature-related features rather than treating backup time as a fixed transformation of nominal capacity; the project targets a software-level backup-time estimate derived from the committed public datasets, rather than reproducing the exact physics-ML architecture of Tu et al. (2024).

5.1.5 Explainable machine learning

Explainability matters when predictions guide maintenance or replacement decisions. SHAP provides a systematic, additive feature-attribution method for explaining individual model predictions, and Tree SHAP specifically is designed to explain tree-based ensemble models such as Random Forest (Lundberg & Lee, 2017; SHAP TreeExplainer documentation, n.d.). Recent battery-specific work stresses the importance of transparent, transferable explanations across chemistries and protocols (Rahmanian et al., 2024). The framework therefore provides both global and local explanations and states explicitly that feature attribution reflects model influence, not physical causation.

5.1.6 Commercial and manufacturer battery-management systems

Commercial systems can provide monitoring and alarms at scale, but their algorithms and data formats are often proprietary and do not expose model assumptions, prediction logic, or reusable outputs (Kurucan et al., 2024). This motivates an open, reproducible, educational prototype rather than a closed vendor tool.

Table 1. Comparison of existing approaches to battery health and runtime estimation.

Approach	Strengths	Limitations	Implication for this project
Formula-based runtime	Transparent, cheap, easy to verify	Assumes stable load; ignores nonlinear fade, thermal effects, cycle history	Retained as a benchmark, not the final estimator
Coulomb counting / filter / equivalent-circuit	Can represent temporal state and remaining discharge time; widely deployed	Drifts over time; needs model structure, calibration, and parameter/state assumptions	Evidence that runtime is predictable, but not the selected core implementation
ML SOH / capacity estimation	Captures nonlinear degradation; higher accuracy with engineered features	Data-hungry; can be a black box; limited cross-chemistry generalisation	Used as the health-assessment branch (leakage-safe)
ML remaining energy / runtime	Accounts for load/C-rate and electro-thermal behaviour	Needs representative dynamic operating data and well-defined targets	Used as the dynamic backup-time branch
Explainable ML (SHAP / feature importance)	Shows which inputs drive a prediction; improves trust	Attribution is not causation; correlated features can distort interpretation	Integrated with both evaluation and dashboard outputs

Approach	Strengths	Limitations	Implication for this project
Commercial / vendor BMS	Operationally mature, integrated telemetry and alarms	Proprietary algorithms and formats; not reusable for research	Motivates an open, reproducible educational prototype

5.2 Research Gap

The literature reviewed above shows that battery State of Health and capacity estimation is already a mature, well-studied area (Berecibar et al., 2016; Roman et al., 2021; Ren & Du, 2023; Khaleghi et al., 2024), that remaining discharge time and energy can be estimated under varying operating conditions (Wang & Chen, 2020; Tu et al., 2024), and that explainability methods such as SHAP are well developed and increasingly applied to battery data (Lundberg & Lee, 2017; Rahmanian et al., 2024). Framed honestly, then, the gap this project addresses is one of integration and reproducibility rather than a new predictive concept. No open, reproducible tool currently combines health assessment, load-aware runtime prediction, a transparent engineering baseline, explanation of individual predictions, and evaluation on independent public datasets within one application-focused, decision-support framework for stationary inverter and backup-power systems.

For planning purposes the gap is split by tier. The first set below is addressed within this Study Project (Phases 2-3); the second is named explicitly as the wider gap this project transitions into for a possible future capstone stage.

Study Project gap: no existing open tool combines health estimation, load-aware runtime prediction, transparent explainability, and an actionable recommendation in one reproducible framework validated against a formula baseline and against independent public datasets.

Capstone/thesis gap: self-supervised or label-efficient learning for real deployments that lack ground-truth capacity labels, a learned (rather than rule-based) recommendation engine, real field validation, embedded/edge deployment, and digital-twin or cloud integration.

- Lack of integration between health assessment and dynamic runtime prediction.
- Limited comparison between formula-based and machine-learning estimates on the same evaluation records.
- Insufficient explanation of individual battery predictions in most deployed tools.
- Lack of transparent, end-to-end, reproducible educational prototypes.
- Difficulty adapting proprietary systems to new datasets or applications.
- Limited support for dataset validation and data-quality reporting.
- Need for a dashboard that connects model development with operational use.
- Absence of a public benchmark dataset for lead-acid household-inverter, UPS, or telecom batteries. This is a gap in the wider literature, not specific to this project (Jiang & Song, 2022).
- Reliance of most existing SOH/RUL models on labelled capacity data that real field deployments rarely have, since true capacity is normally confirmed only by a full discharge test.
- Limited development, in the wider literature, of a decision-support or recommendation layer that turns a health/runtime prediction into an actionable maintenance decision rather than stopping at a number.

5.3 Dataset Documentation

This project uses both primary and secondary datasets across its three development phases. A primary dataset is one generated or collected directly by the project team; a secondary dataset is an existing dataset collected by an external institution and obtained through a public research repository. This classification is based on who collected the data, not on how important it is to model training. The table below lists every dataset committed to the project, its classification, the phase(s) in which it is used, and its main purpose. Phase 1 work is limited to identification, access verification, schema mapping, and feasibility assessment: no model training has been carried out at this stage.

ID	Dataset	Classification	Phase(s)	Main Purpose
P1	Project-Generated Synthetic Battery Dataset	Primary	1, 2, 3	validation/error-case testing, pipeline testing (not used for accuracy or any evaluation at all)
S1	Second-Life Lithium-Ion Battery Aging Dataset (Stanford)	Secondary: Core	1, 2, 3	Principal real-world dataset for SOH/capacity modelling and stationary-storage duty-cycle behaviour
S2	NASA Li-ion Battery Aging Dataset (PCoE)	Secondary: Core	1, 2, 3	Independent capacity/SOH validation under controlled charge-discharge-EIS protocols
S3	NASA Randomized Battery Usage 2 (Room-Temperature Random Walk)	Secondary: Core	1, 2, 3	Variable-load / generalisation testing under unscheduled current
S4	Oxford Energy Trading Battery Degradation Dataset	Secondary: Core	1, 2, 3	Dynamic, real-world-like stationary-storage load validation
S5	CALCE Battery Data (University of Maryland)	Secondary: Supporting	3	Cross-dataset / cross-form-factor evaluation using a schema-compatible subset
S6	Oxford Battery Degradation Dataset 1 (Artemis urban profile)	Secondary: Optional	3	Supplementary dynamic-discharge validation; useful for generalisation but a drive-cycle load pattern rather than the project's stationary backup-load focus
S7	NASA Randomized and Recommissioned Battery Dataset	Secondary: Optional	3	Optional pack-level / second-life robustness study
S8	Severson et al. Fast-Charge LFP Dataset	Secondary: Optional	3	Optional cross-chemistry generalisation test

Dataset Descriptions and References

P1: Project-Generated Synthetic Battery Dataset (Primary). Generated using the project team's own Python scripts for software development and functional validation, including deliberately invalid and missing-value records used to test the validation and cleaning modules. Main variables: voltage, current, temperature, capacity, rated capacity, load, cycle

number, and backup time. This data verifies only that the full pipeline (import → validation → cleaning → feature engineering → ML → dashboard) executes correctly; it is not used to support real-world accuracy claims.

S1: Second-Life Lithium-Ion Battery Aging Dataset (Secondary, Core). Six previously aged INR21700-M50T cells with NMC cathodes and graphite/silicon anodes, cycled at the Stanford Energy Control Laboratory using synthetic residential and commercial grid-storage duty cycles at 20°C and 35°C, with periodic capacity, pulse-power, and EIS diagnostic tests. This is the project's principal real-world dataset for capacity/SOH modelling and stationary-storage load behaviour.

Moy, K., Khan, M. A., Fasolato, S., Pozzato, G., Allam, A., & Onori, S. (2024). Second-life lithium-ion battery aging dataset based on grid storage cycling [Data set]. Open Science Framework. https://doi.org/10.17605/OSF.IO/8JNR5; and Data in Brief, 57, 111046. https://doi.org/10.1016/j.dib.2024.111046

S2: NASA Li-ion Battery Aging Dataset (Secondary, Core). Commercial 18650 Li-ion cells subjected to repeated charge, discharge, and electrochemical impedance spectroscopy (EIS) cycles at different temperatures and current loads at the NASA Ames PCoE, continued until approximately 30% capacity fade. The same underlying PCoE collection is referenced both through its original creators and through the public data.gov catalogue entry; both citations point to the same experimental source and are listed together below to avoid double-counting it as two datasets.

Saha, B., & Goebel, K. (2007). Battery data set [Data set]. NASA Prognostics Data Repository, NASA Ames Research Center; and National Aeronautics and Space Administration. (2010). Li-ion battery aging datasets [Data set]. https://catalog.data.gov/dataset/li-ion-battery-aging-datasets

S3: NASA Randomized Battery Usage 2 (Secondary, Core). Four 18650 Li-ion cells (RW3-RW6) continuously cycled using a randomized sequence of discharge currents between 0.5 A and 4 A, with reference charge/discharge cycles performed after every 50 random-walk cycles to provide SOH benchmarks. Used to test robustness under variable, unscheduled load.

National Aeronautics and Space Administration. (2022). Randomized battery usage 2: Room temperature random walk [Data set]. NASA Open Data Portal. https://data.nasa.gov/dataset/randomized-battery-usage-2-room-temperature-random-walk

S4: Oxford Energy Trading Battery Degradation Dataset (Secondary, Core). Approximately one year of experimental measurements from six lithium-ion cells operated with variable current profiles designed for grid-connected batteries participating in day-ahead electricity trading, including current, voltage, temperature, and monthly capacity readings. Chosen as the core Oxford dataset because its dynamic, grid-storage load pattern is the closest public analogue to a residential or commercial backup-storage duty cycle, which matches this project's inverter/UPS application focus.

Reniers, J. M., Mulder, G., & Howey, D. A. (2020). Oxford energy trading battery degradation dataset [Data set]. University of Oxford. https://doi.org/10.5287/bodleian:gJPdDzvP4

S5: CALCE Battery Data (Secondary, Supporting). Open experimental Li-ion data from the University of Maryland's Center for Advanced Life Cycle Engineering, including full/partial cycling, storage, dynamic driving profiles, OCV, and impedance measurements across multiple form factors and chemistries. A schema-compatible subset is used for cross-dataset evaluation and to test the pipeline's portability beyond the core datasets above.

CALCE Battery Research Group. (n.d.). Battery data [Data set]. University of Maryland. https://calce.umd.edu/battery-data

S6: Oxford Battery Degradation Dataset 1 (Secondary, Optional). Eight lithium-ion pouch cells degraded under a dynamic Artemis urban driving profile, with long-term degradation measurements. Kept as an optional, supplementary dynamic-load validation set: it broadens coverage of dynamic discharge behaviour, but its drive-cycle current profile is a less direct match to stationary backup-power applications than S4, so it is not treated as a core dataset.

Birkl, C. R. (2017). Oxford battery degradation dataset 1 [Data set]. Oxford University Research Archive. https://doi.org/10.5287/bodleian:KO2kdmYGg

S7: NASA Randomized and Recommissioned Battery Dataset (Secondary, Optional: Phase 3). Twenty-six two-cell battery packs tested under constant and randomized loads, including second-life packs assembled from surviving cells of previously aged packs. Kept optional, for pack-level and second-life robustness testing if time permits in Phase 3.

National Aeronautics and Space Administration. (2023). Randomized and recommissioned battery dataset [Data set]. NASA Open Data Portal. https://data.nasa.gov/dataset/randomized-and-recommissioned-battery-dataset

S8: Severson et al. Fast-Charge LFP Dataset (Secondary, Optional: Phase 3). 124 A123 LFP 18650 cells cycled under 72 different fast-charging protocols at MIT/Stanford/Toyota Research Institute. Kept as an optional cross-chemistry generalisation test, not required for the core Study Project objectives.

Severson, K. A., Attia, P. M., Jin, N., Perkins, N., Jiang, B., Yang, Z., Chen, M. H., Aykol, M., Herring, P. K., Fraggedakis, D., Bazant, M. Z., Harris, S. J., Chueh, W. C., & Braatz, R. D. (2019). Data-driven prediction of battery cycle life before capacity degradation. Nature Energy, 4, 383-391. https://doi.org/10.1038/s41560-019-0356-8

Target Construction and Dataset Compatibility

Not every committed dataset contains a column explicitly named “backup time.” Where technically valid, the backup-time target is derived from the measured discharge timeline: the remaining time from a defined observation point to the dataset's discharge cut-off condition. Energy and load-related features are similarly derived from time-aligned voltage and current measurements. Different chemistries, experimental protocols, and target definitions across datasets are not simply concatenated into one table; each dataset is mapped through its own adapter into a common schema before comparison, and target construction is documented per dataset so that it is reproducible and does not leak future information into the model inputs.

Justification and Limitation

No public benchmark dataset exists for lead-acid batteries in household-inverter, UPS, or telecom-backup applications; this was checked across Kaggle, Mendeley Data, Zenodo, UCI, OpenML, and the major battery-research repositories, and is consistent with the wider literature (Jiang & Song, 2022). Building a lead-acid-specific dataset would require a multi-week degradation experiment outside the equipment and time available for this Study Project. The committed datasets were instead chosen to be complementary and reproducible: the second-life dataset supplies duty-cycle-realistic residential/commercial load profiles directly relevant to the inverter/UPS application; the NASA datasets add controlled-protocol and randomized-load ageing regimes; the Oxford energy-trading dataset adds dynamic, real-world-like load behaviour; and CALCE adds cross-form-factor coverage. The software framework, not the dataset choice, is what is intended to generalise: the import → validation → cleaning → feature engineering → prediction → explanation → recommendation → dashboard pipeline is designed to be chemistry- and domain-agnostic, so extending it to a new battery type or application is a data-schema change, not an architecture change.

This prototype is trained and evaluated exclusively on public Li-ion laboratory and second-life datasets. It has not been, and at this phase cannot be, validated against real inverter, UPS, or telecom field telemetry, nor against lead-acid or other non-Li-ion chemistries. The claim that the framework generalises across battery-powered applications is an architectural design claim at this stage, not an empirically demonstrated one.

5.4 Functional Modules and Architecture

The proposed system is composed of ten interconnected functional modules that transform raw battery data into explainable health and backup-time predictions. Modules M1-M3 build a validated, leakage-safe feature layer shared by every downstream branch. M4 (the formula baseline), M5 (capacity/SOH), and M6 (backup-time ML) run in parallel on that shared feature layer so that the baseline and the machine-learning models are compared on equal terms rather than treating the baseline as an afterthought. Their outputs converge at M7 (evaluation), feed M8 (explainability) and M9 (decision support), and are presented through M10 (the dashboard). M9 is a secondary, non-safety-critical enhancement and is not part of any certified maintenance or safety decision.

No.	Module	Purpose	Key Inputs	Key Outputs
M1	Dataset Ingestion	Load supported CSV/Excel files and adapt source-specific fields to a common schema	Raw files from committed datasets or device logs	Raw, standardized battery records
M2	Validation & Preprocessing	Check data quality; handle missing values, duplicates, and out-of-range records	Raw standardized records	Validated, cleaned dataset + quality report
M3	Feature Engineering	Generate leakage-safe electrical, thermal, cycle, load, and energy features	Cleaned data	Model-ready feature matrices (e.g., C-rate, ΔV, internal resistance, energy)
M4	Formula-Based Baseline	Compute a transparent, configurable energy/load-based backup-time estimate	Voltage, capacity, usable fraction, load, efficiency	Baseline backup-time estimate
M5	Capacity / SOH Assessment	Estimate remaining capacity and derive SOH relative to rated capacity	Engineered features	Predicted capacity, SOH (%)
M6	Dynamic Backup-Time ML	Predict backup time under the current load using Random Forest	Engineered features, SOH, current load, voltage	Predicted backup time (minutes)
M7	Evaluation & Comparison	Compare actual, formula, and ML estimates using multiple metrics	Actual runtime; predicted runtime (formula & ML)	MAE, RMSE, MAPE, bias, R^2, plots, residuals
M8	Explainability	Explain global feature importance and individual predictions	Trained model, engineered features	Feature-importance plots, SHAP summary & waterfall plots
M9	Rule-Based Decision Support (secondary)	Map SOH, backup time, load, and data-quality indicators to configurable,	M5-M8 outputs	Recommendation message + supporting rationale

No.	Module	Purpose	Key Inputs	Key Outputs
		non-safety-critical guidance		
M10	Dashboard & Reporting	Provide interactive visualisations, inputs, alerts, and downloadable outputs	All module outputs	Interactive dashboard, reports, exports

5.4.1 Illustrative Recommendation Rules (Preliminary)

The rule-based recommendation layer (Module M9) maps predicted State of Health to a maintenance recommendation as follows:

SOH Range	Recommendation
90% and above	No action needed
80% to 89%	Monitor more frequently
65% to 79%	Reduce load or schedule inspection
Below 65%	Plan replacement

Independent of the SOH band above, if the predicted backup time falls below the load's required runtime, the recommendation escalates to "reduce load" or "plan replacement" regardless of the SOH-based category. These thresholds are preliminary. They will be calibrated in Phase 2 and Phase 3 against real capacity-fade data from the committed datasets (Section 5.3), rather than being finalised on assumption alone.

5.5 Functional Requirements

FR1. Import supported CSV or Excel battery datasets.

FR2. Provide a bundled/synthetic demonstration dataset when no external dataset is supplied, for schema and pipeline testing only.

FR3. Map source-specific column names to a documented common schema.

FR4. Verify that required fields, data types, and units exist and are physically plausible before training or prediction.

FR5. Convert battery measurements to numeric values and report missing, invalid, duplicate, and out-of-range records.

FR6. Apply documented cleaning and preprocessing rules to accepted records.

FR7. Generate leakage-safe discharge-cycle, load, thermal, and energy-related features (e.g., C-rate, ΔV, internal resistance, delivered energy).

FR8. Derive a discharge-duration/backup-time target from time-series discharge records where the source data support a valid, leakage-safe derivation.

FR9. Calculate a configurable, formula-based backup-time baseline.

FR10. Train a Random Forest regression model for dynamic backup-time prediction.

FR11. Train a separate, leakage-safe regression model for remaining-capacity estimation.

FR12. Calculate predicted and actual State of Health from predicted/actual and rated capacity.

FR13. Classify battery health into configurable categories.

FR14. Compare formula-based and machine-learning predictions on the same evaluation records.

FR15. Calculate MAE, RMSE, MAPE, bias, and R^2 where mathematically valid.

FR16. Display actual-versus-predicted and residual/error visualisations.

FR17. Display global feature-importance and permutation-importance charts.

FR18. Generate local (individual-prediction) explanations and SHAP charts when the SHAP package supports the selected model.

FR19. Accept a new battery operating condition through dashboard inputs and return capacity, SOH, health class, formula runtime, and ML runtime.

FR20. Map predicted SOH, backup time, load, and data-quality indicators to configurable, non-safety-critical maintenance recommendations, following the preliminary thresholds defined in Section 5.4.1.

FR21. Display each recommendation alongside its supporting feature-importance/SHAP rationale.

FR22. Export cleaned data, predictions, metrics, and explanation outputs.

FR23. Save trained models and preprocessing artefacts for later reuse without mandatory retraining.

FR24. Display assumptions and limitations so that demonstration outputs are not mistaken for certified, safety-validated, or field-validated results.

5.6 Non-Functional Requirements

Quality Attribute	Requirement
Usability	Clear labels and units; validation warnings separated from fatal errors; normal use should not require editing source code; results presented through metrics, tables, and charts.
Performance	Validation and prediction remain interactive for small/medium datasets; model training is practical on a normal student laptop; saved models support reuse without mandatory retraining.

Quality Attribute	Requirement
Reliability	Invalid data must not silently enter training; fixed random seeds and documented splits support reproducibility; preprocessing is identical between training and prediction.
Security	Uploaded files are treated as data, not executable content; file types and schemas are validated; production deployment would add authentication, access control, size limits, secure storage, and audit logging.
Maintainability	Code is divided into logical modules; thresholds/configuration are centralised; dependencies are declared; features, targets, formulas, and assumptions are documented; version control is used throughout.
Scalability	The design permits additional algorithms, datasets, chemistries, and a persistent data store in later phases; large time-series datasets may require batching or database-backed processing.
Explainability	At least one global and one local explanation are provided; the interface states that model importance is not proof of causation and that correlated features require cautious interpretation.
Portability	The core prototype runs in a documented Python environment on a standard laptop and does not require a GPU for the initial Random Forest implementation.

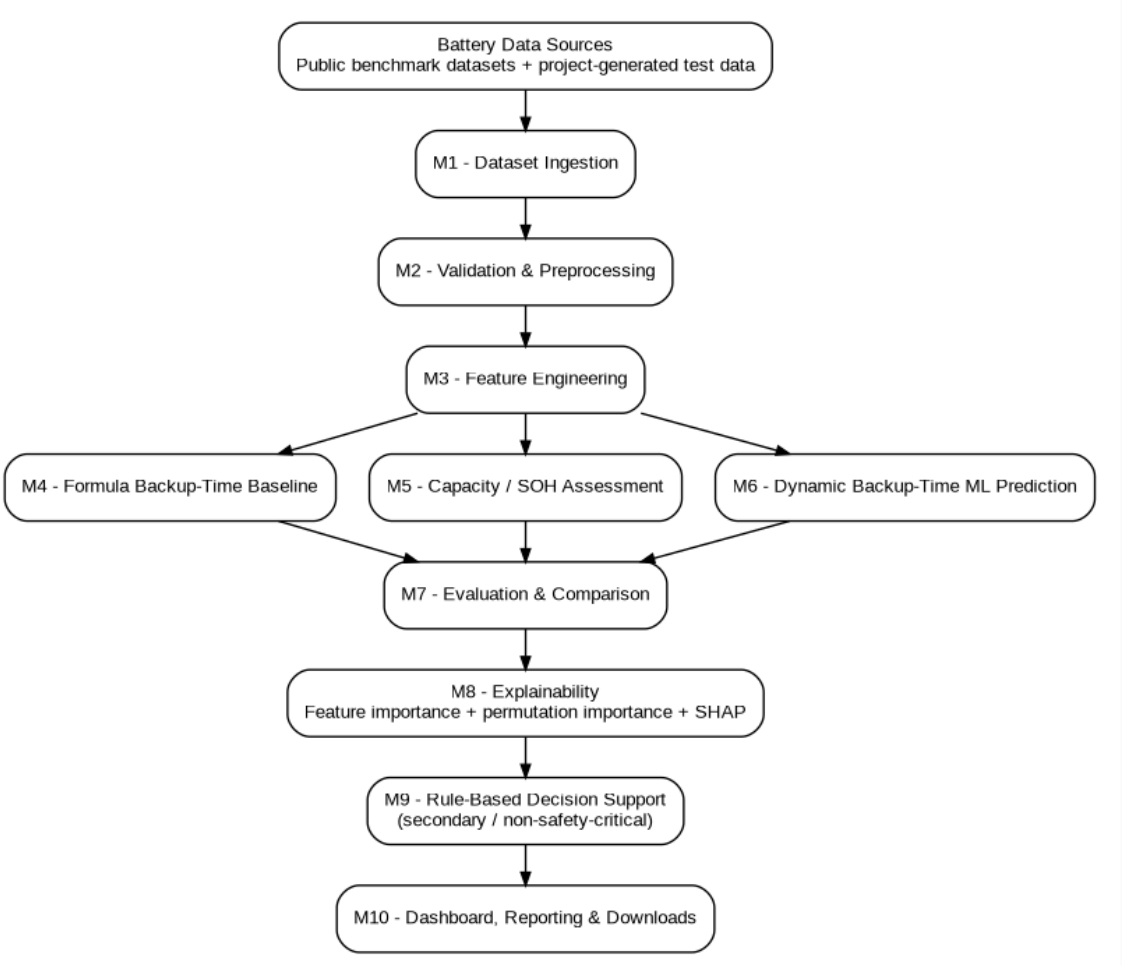


Figure 1. High-level functional workflow and module interactions.

5.7 Feasibility Analysis

Technical feasibility. The required tools are mature and accessible: Python; pandas and NumPy for data processing; scikit-learn for Random Forest modelling and evaluation; Matplotlib for plots; SHAP for explainability; Streamlit for the dashboard; Joblib for model persistence; and GitHub for collaboration and version control. These are tools the team already has working familiarity with. The main technical challenge is not tool availability but harmonising heterogeneous datasets without introducing target leakage or overstating cross-domain generalisation; Phase 1 has been limited to planning, with implementation scheduled to begin in Phase 2.

Time feasibility. The project is staged across the official three-phase academic structure and a set of incremental milestones: requirements and research; dataset schema and validation; preprocessing and feature engineering; the formula baseline; machine-learning models; evaluation; explainability; dashboard integration; and testing and final documentation. The initial prototype prioritises one application domain and one main regression algorithm; advanced optimisation remains optional. Exact calendar dates for later sub-phases will follow the official course schedule rather than being fixed unilaterally in this document (see Section 7).

Resource feasibility. The project runs on a normal laptop and does not require a GPU. The main resource constraint is access to a sufficiently representative battery dataset; the committed public datasets (Section 5.3) support core research and development, while synthetic data validates software execution only. Final performance claims require the independent, held-out validation described in Section 5.3. The absence of project-collected field telemetry is the main resource gap, and it is treated as an explicit limitation rather than assumed away.

Validation protocol. Model evaluation will use leave-one-battery-out (LOBO) cross-validation as the primary protocol: in each fold, all cycles from one battery are held out as the test set while the model trains on the remaining batteries, repeated so that every battery is tested exactly once. This avoids the leakage that a random row-level split would cause, since consecutive cycles from the same battery are highly correlated. A secondary time-based split (earlier cycles for training, later cycles of the same battery for testing) will also be run, but only as a sanity check on temporal generalisation, not as the primary reported result.

6. Project Scope and Expected Deliverables

6.1 Included Scope

- A software prototype focused on battery health and backup-time prediction for stationary inverter/UPS-style applications.
- Battery data containing voltage, current, temperature, capacity, load, cycle number, and measured or derived backup time.
- CSV and Excel import, with source-specific schema adapters.
- Dataset validation, cleaning, and leakage-safe feature engineering.
- A formula-based backup-time baseline with configurable efficiency/usable-capacity assumptions.

- Random Forest as the core regression algorithm, with a separate leakage-safe capacity/SOH workflow.
- Model evaluation, feature importance, permutation importance, and SHAP explanations.
- An interactive dashboard with plots, metrics, explanations, validation feedback, and downloadable outputs.
- A rule-based, non-safety-critical maintenance-recommendation layer, as a secondary enhancement once core modules are complete.
- Source code, technical documentation, and saved model/preprocessing artefacts.

6.2 Excluded from Scope

- Design or manufacture of battery hardware; direct control of chargers, inverters, or EV systems; cell balancing; charging optimisation; thermal-control actuation.
- Real-time embedded BMS deployment, safety certification, fault isolation for every failure mode, or production-grade cloud deployment.
- Remaining Useful Life (RUL) prediction as a core target; RUL may be considered only as a future enhancement after the capacity/SOH and backup-time objectives are validated.
- Claims of validated performance on lead-acid, UPS/inverter field, or telecom data unless such data is independently obtained and tested.
- Claims that the model is chemistry-agnostic; the architecture is extensible, but new chemistries require retraining, calibration, and validation.
- Use of synthetic data as evidence of real-world predictive accuracy.
- Automated data collection from physical sensors.

6.3 Assumptions and Constraints

Type	Statement
Assumption	Input measurements have documented units and sufficient quality for the selected target and features.
Assumption	Rated/reference capacity is known or can be defined consistently for SOH calculation.
Assumption	A valid discharge cut-off and time sequence exist when a backup-time target must be derived from source data.
Assumption	Formula parameters such as efficiency and usable-capacity fraction are configurable, documented baseline assumptions.
Constraint	Public datasets differ in chemistry, format, protocol, and operating conditions; not every field can be shared across all sources.
Constraint	Cross-dataset generalisation may be limited by chemistry, protocol, temperature, sensor, and target-definition differences.
Constraint	Feature importance can be distorted by correlated engineered variables and should not be interpreted as causal evidence.
Constraint	Time available across the academic phases limits the number of algorithms, tuning experiments, and deployment environments that can be validated.

6.4 High-Level System Overview

The proposed workflow (Figure 1, Section 5.4) is: committed battery datasets → import and schema validation → cleaning and preprocessing → discharge-cycle and load feature engineering; the formula baseline and the machine-learning models (capacity/SOH and backup time) then run in parallel on the same engineered features; their outputs feed performance comparison, feature importance, and SHAP explanation; and everything is surfaced through the interactive dashboard and downloadable exports.

6.5 Expected Deliverables

Phase 1 deliverables (this document)

- Project idea summary, finalised title, and abstract with a literature-supported research gap.
- Problem statement, motivation, educational value, and course alignment.
- Primary and secondary objectives with a clear core/enhancement boundary.
- Literature review and a dedicated Research Gap section with APA in-text citations.
- Dataset documentation with Primary/Secondary classification, justification, and references.
- Functional module definitions, functional/non-functional requirements, and a high-level workflow diagram.
- Feasibility analysis, scope definition, assumptions, constraints, risk register, and phase-level timeline.
- GitHub repository structure and a complete APA 7th-edition reference list.

Later-phase deliverables

- Phase 2: finalised common data schema and source adapters; detailed architecture and data-flow design; implemented ingestion/validation/preprocessing and feature-engineering modules; implemented formula baseline and an initial ML slice; a working end-to-end Proof-of-Concept demo; defined testing strategy.
- Phase 3: trained backup-time and capacity/SOH models; battery-level cross-validation and independent held-out evaluation across the committed datasets; explainability and recommendation modules; a working Streamlit dashboard; saved trained models and generated CSV/JSON/chart outputs; test evidence and screenshots; final technical report, demonstration, and complete source-code archive.

7. Preliminary Project Timeline and Milestones

This Study Project follows the official three-phase schedule. The table below gives the academic-stage overview; exact calendar dates for Phase 2 and Phase 3 sub-activities will be finalised against the official course schedule rather than fixed unilaterally here. The working sub-phase plan beneath it (originally targeting 20 Jul - 20 Sep 2026) is retained as a planning aid so that weekly progress can still be tracked against phase-level deliverables, and will be re-dated once the official schedule is confirmed.

7.1 Academic-Stage Overview

Academic Stage	Main Activities	Milestone / Output
Semester 5 - Phase 1: Problem Identification & Planning	Finalise title; literature review; research gap; datasets; objectives; functional modules; workflow; requirements; scope; feasibility; risks; plan.	Approved Phase-1 proposal (this document).
Semester 5 - Phase 2: Design & Proof of Concept	Finalise schema and source adapters; detailed architecture and data flow; implement ingestion/validation/preprocessing; implement formula baseline and initial ML slice; demonstrate PoC; define testing strategy.	Detailed design, dataset pipeline, architecture, and working PoC.
Semester 6 - Phase 3: Implementation Readiness & Validation	Complete and integrate ML health/runtime modules; execute held-out/cross-dataset evaluation; implement explainability, recommendations, and dashboard; run test cases; review performance, risks, and limitations; finalise report/demo.	Validated final implementation and submission package.

7.2 Working Sub-Phase Plan (indicative dates, pending official confirmation)

Sub-Phase	Indicative Window	Activities → Deliverable
1.1 Project initiation	Week 1	Confirm domain, title, problem statement, stakeholders, expected outputs → approved project idea.
1.2 Background research & literature review	Week 2	Review SOH/runtime literature, datasets, ML approaches, explainability, and decision-support gaps; build the cited bibliography → literature review complete.
1.3 Phase 1 proposal finalisation	Week 3	Finalise objectives, requirements, scope, feasibility, dataset commitment, and risk register; submit the Phase 1 document → Phase 1 proposal submitted.
2.1 Architecture & schema design	Week 4	Finalise the dataset schema and source adapters; produce the architecture diagram, module-wise design, and technology-stack justification → architecture diagram and module design.
2.2 Data pipeline implementation	Week 5	Implement import, validation, cleaning, and feature-engineering modules; document data flow → validated dataset and feature matrix.
2.3 Formula baseline & PoC demo	Week 6	Implement the formula-based backup-time baseline; assemble an end-to-end PoC slice (import → clean → feature → baseline output) → working PoC demo.
3.1 Core ML models	Weeks 7-8	Train the backup-time Random Forest model and the leakage-safe capacity/SOH model; compute SOH and compare against the formula baseline → two trained models plus comparison results.
3.2 Explainability, recommendation & dashboard	Weeks 9-10	Implement feature importance, permutation importance, and SHAP; implement the rule-based recommendation module; integrate everything into the dashboard → working explainable dashboard with recommendations.
3.3 Testing, risk review & final submission	Weeks 11-12	Test edge cases; finalise the risk-mitigation review; document future-work scope; complete the final report and presentation → final report, presentation, and repository.

7.3 Milestones

- M1: Final project title, domain, and problem definition approved.
- M2: Literature review and citation-supported research gap completed.
- M3: Core and optional datasets identified, documented, and classified as Primary/Secondary.
- M4: Primary/secondary objectives and functional modules approved.
- M5: Phase-1 requirements, scope, feasibility, and risk register completed; Phase 1 proposal submitted.
- M6: Phase-2 architecture, schema, and Proof-of-Concept plan completed.
- M7: Core capacity/SOH and backup-time models implemented and evaluated.
- M8: Explainability, recommendation, and dashboard integration completed.
- M9: Phase-3 validation, documentation, and final demonstration completed.

7.4 Scope Boundary: Study Project vs. Future Capstone/Thesis

This Study Project is planned to deliver a working, explainable prediction-and-recommendation prototype on public Li-ion laboratory and second-life data within the academic timeline above. The following items are deliberately out of scope here and are named as possible future capstone/thesis work, so the boundary is explicit rather than implied:

- A learned/ranked recommendation engine, beyond the fixed-rule module planned for Phase 3.
- Self-supervised or weak-label learning to address the label scarcity that real field deployments face; this Study Project trains on fully labelled lab-benchmark data only.
- Validation against real inverter, UPS, or telecom field telemetry, or against lead-acid or other non-Li-ion chemistries.
- Embedded/edge deployment of the trained models.
- Digital-twin or cloud-edge integration.

8. Team Structure and Collaboration

8.1 Team and Workstreams

This Study Project is submitted as a 3-person team. Each member contributes across research, dataset preparation, implementation, testing, review, and documentation; specific task-level ownership within each workstream is tracked through the shared task tracker and GitHub issues rather than fixed permanently in this document, since workload will shift across phases.

Student	ID	Primary Workstream Focus
Dhruv Purohit	2023EB03173	Scope, architecture, integration, and review
Himanshu Kadia	2023EB03114	Literature, datasets, validation, and feature engineering
Nitin Kumar	2023EB03248	Models, evaluation, and explainability

In addition to the primary focus above, dashboard/user-interaction/deployment and testing/reports/presentation/repository quality are shared workstreams that all three members contribute to, with ownership rotating as needed and recorded in the task tracker.

8.2 Collaboration Tools

- GitHub: source code, branches, pull requests, issues, releases, and version history.
- GitHub Projects (or an equivalent Kanban board): milestones, task ownership, priorities, and status.
- Shared documents / Google Drive: proposal drafts, literature notes, meeting records, dataset copies, and presentation files.
- WhatsApp, Zoom, and email: team discussion and supervisor coordination.
- README, requirements.txt, and configuration files: reproducible setup, dependencies, current limitations, and run instructions.

9. Risk and Challenge Analysis

9.1 Risk Register

Risk	Prob.	Impact	Mitigation Strategy
Insufficient real battery data	High	High	Use the committed public datasets (Section 5.3) for development; define a minimum schema; use synthetic data only for software demonstration.
Dataset inconsistency / missing fields	High	High	Use source-specific adapters; standardise units and column names; validate ranges; document target construction; reject unusable records.
Synthetic results appear unrealistically strong	High	High	Label all synthetic results clearly; avoid final accuracy claims from synthetic data; repeat evaluation on independent real data.
Target leakage	Medium	High	Separate health and backup feature sets; exclude capacity-derived or future-derived features; review pipelines before training.
Model overfitting	Medium	High	Use held-out test data, out-of-bag score, cross-validation where possible, limited model complexity, and independent cross-dataset checks.
Poor generalisation across chemistry/protocol	High	High	Record chemistry/protocol metadata; avoid chemistry-agnostic claims; report per-dataset performance; retrain/calibrate for new domains.
Correlated engineered features distort importance	High	Medium	Use impurity, permutation, and SHAP together; explain correlation; avoid causal claims.
Missing software dependencies / environment problems	Medium	Medium	Provide requirements.txt, virtual-environment instructions, launch scripts, and dependency checks.
Dashboard deployment problems	Medium	Medium	Test locally first; use a standard Streamlit command; provide sample data and fallback modes.
Excessive project scope / time pressure	Medium	High	Prioritise one battery domain, one main algorithm, and essential dashboard features; freeze advanced

Risk	Prob.	Impact	Mitigation Strategy
			features before final testing; defer RUL, field deployment, and learned recommendation engines.
Inaccurate formula assumptions	Medium	Medium	Keep efficiency and usable-capacity values configurable; document that the formula is a baseline, not ground truth.
Incorrect SOH thresholds	Medium	Medium	Treat thresholds as configurable demonstration settings; adapt them to chemistry and application guidance.
Sensor noise or missing values	High	Medium	Use validation, median imputation, outlier reporting, robust features, and data-quality indicators.
Security issues with uploaded files	Low locally / higher in deployment	High	Restrict file types and size; parse rather than execute files; add authentication and isolation for production deployment.
Prototype treated as a safety decision without validation	Low	High	Display limitations; require expert review; state explicitly that the prototype is decision support, not a certified safety system.
Loss of work or incompatible versions	Medium	Medium	Use GitHub, tagged releases, backups, locked dependencies where practical, and archived project packages.
No ground-truth labels from field deployments	High	Medium	Restrict quantitative accuracy claims to lab-benchmark test splits; treat any future field telemetry as an unlabeled distribution-shift sanity check only, not a scored validation set.

9.2 Main Challenge Areas

Data availability and comparability. Battery degradation experiments are long, and public datasets differ in temperature, cut-off voltage, chemistry, sampling rate, and cycling protocol. Mitigation: standardise only what can be justified, begin with the committed public datasets and a well-defined schema, keep dataset-specific metadata, and report incompatible records rather than forcing them into one model.

Dynamic operating conditions. Backup time changes with load, temperature, capacity, voltage response, and the chosen discharge cut-off. Mitigation: include load, energy, C-rate, thermal, and cycle-stress features; compare predictions under different loads; extend to time-series data in later phases.

Explainability versus accuracy. A model with strong predictive performance is not automatically understandable. Mitigation: begin with Random Forest; combine impurity-based feature importance, permutation importance, and SHAP; present individual prediction explanations; explicitly acknowledge correlation and non-causality.

Reproducibility. Differences in environments, datasets, and random splits can change results. Mitigation: fixed random seeds, saved configuration files, declared dependencies, exported metrics, and persisted model packages.

Maintaining a realistic scope. Battery science broadly includes SOC, SOH, RUL, thermal safety, fault diagnosis, and charging control. Mitigation: limit the core prototype to remaining

capacity, SOH, and backup-time prediction, and treat other capabilities as future extensions (Section 7.4).

10. Preliminary Conclusion

The proposed Study Project is technically feasible, educationally valuable, and suitable for development within the available timeline. It addresses a meaningful battery-management problem by combining a transparent formula calculation with adaptable machine-learning predictions and model explanations, evaluated across a deliberately diversified set of independent public datasets (second-life, NASA, Oxford, and CALCE) rather than a single source.

The project is differentiated from a simple data-analysis exercise because it plans a complete software workflow: dataset import, validation, preprocessing, feature engineering, baseline calculation, machine-learning models, evaluation, explainability, saved artefacts, and an interactive dashboard. As the literature review in Sections 5.1-5.2 makes clear, the team's own view is that the project's main contribution is the integration of established individual methods into one reproducible, explainable, and load-aware framework, rather than a new predictive technique.

The most important requirement for final project credibility is an honest separation between implementation validation, within-dataset validation, and independent real-data validation (Section 5.3). Synthetic data can demonstrate that the software functions correctly, but the final report must keep these three forms of evidence clearly distinguished, and must not extend performance claims beyond the chemistries and operating conditions actually represented in the committed datasets.

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All references follow APA 7th-edition format and are listed alphabetically by author. Every entry below is cited at least once in the body of this document.

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11. Supervisor Review and Approval

Advisor Feedback:

Supervisor Comments:

Recommendations:

Signature: _____

Date: _____